A Multi-Modal Machine Learning Approach for Phishing Website Detection Using URL and Visual Features

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Abstract—Phishing attacks are one of the most significant threats in the area of cyber security and have increasingly started using traditional methods of passing through detection techniques using intelligence visual simulations. This paper states the development of a new approach that combines the power of Convolutional Neural Networks and Large Language Models for detecting phishing. The work uses the Efficient-B0 network for the automatic extraction of high-level features from website screenshots and also DistilBERT.The proposed system has been implemented and tested using a custom-labeled dataset comprising 2,282 screenshots and more than 10,294 URLs. From the results obtained, the Fusion model is observed to be significantly better than single-modality systems and has produced excellent results with an accuracy of 95.3%, precision of 98%, and F1-score of 95.16%. Analysis of confusion matrices indicates that the hybrid technique has been able to effectively counter weaknesses of individual models by performing well on detection of "zero-day" phishing pages with fresh layout designs or those using disguised links. Though certain limitations are involved in terms of computational complexity and dependability on accurate matching of image URLs, this research claims that it is an enormously strong technique for such purposes.

Keyword— Cybersecurity, Phishing, Deep Learning, CNN,Multimodal Fusion, Large Language Models, DistilBERT, EfficientNet.

I. Introduction

The growing usage of online services including digital banking, electronic commerce, cloud computing, and social networks has raised the potential risks of cyberattacks. Phishing is considered as the most common and harmful online scam that targets consumers by mimicking internet portals in order to gain confidential data such as login passwords and financial information [1]–[3]. Modern phishing websites are extremely clever, and they are typically impossible to identify from official internet portals due to their design, which is unsolvable using traditional approaches.

To address these problems, a lot of research has been done on automatic phishing site detection that combines machine learning (ML) and deep learning (DL) [4]–[6]. A recent study investigates web content structures and visual characteristics using ensemble learning techniques and convolutional neural networks (CNNs), whereas previous research on phishing site detection relied on traditional classifiers with carefully constructed URL-based features [7], [8]. These methods have certain particular drawbacks even though they are successful in achieving high detection rates. The majority of current DL-based methods for this task are typically computationally demanding and unsuitable for real-time situations, while their lightweight counterparts might be less resistant to complex phishing websites [9], [10]. The majority of current efforts on this issue are often centered on a single modality, either URL-based or visual similarity-based, which leaves them open to attackers attempting to circumvent particular feature sets.

Structural analysis of URLs and online pages is made possible by recent developments in the field of Large Language Models (LLMs), which present an appealing answer to the phishing detection task. LLMs are capable of analyzing and identifying context-dependent and language indicators, such as misunderstanding instructions, using inappropriate terminology, and using social engineering techniques, that are frequently employed by attackers when creating phishing pages [3], [11]. Only flaw with the previous strategy is that it ignores the phishing pages' visual deception.

The goal of this work is to use models that are both visually and semantically informed to develop an adaptive and multimodal phishing website detector. This detector is designed to be computationally efficient and less susceptible to visually deceptive phishing attacks. The detector is a binary classifier since it uses a website's URL and related image to identify whether or not it is a phishing website. A phishing website has a predicted class value of 1, while a legal website has a predicted class value of 0. The detector can be realized by combining CNNs and LLMs.

The importance of this study is based on filling multiple existing gaps in conventional anti-phishing methods. Specifically, this proposed solution tries to address both the efficiency and real-time capability in anti-phishing systems. Additionally, this solution tries to increase robustness against changing attack patterns. By exploring multiple sources of information from both visual and semantic perspectives, this solution tries to make anti-phishing methods more efficient and scalable.

Methodologically, screenshots of web pages are analyzed by convolutional Neural Networks to identify key visual cues. On the other hand, large language models are employed to analyze URLs and text to identify semantic patterns of phishing. These results are then combined to reach a final decision. To serve as the basis of a new project, it is recommended that the approach be supplemented by an auto-optimization module that enables the determination of the optimum size of the URL and the optimum size of the webpage screenshot to be used when applying the approach. The rest of this paper is organized into the following sections. Section II introduces the existing works on detecting phishing websites. The suggested approach and system design are further explained in Section III. The performance analysis and results are presented in Section IV. The limitations and possible future research are discussed in Section V, and the paper is concluded in Section VI.

II. Literature Review

Phishing is still one of the most common and dangerous forms of cybercrime that targets users of social media, e-commerce cloud platforms, and online banking. Phishing sites are getting better all the time, and they often copy websites in terms of both structure and style, making it almost impossible for people to tell them apart. As a result, automated methods for finding phishing websites have become a field of study in cybersecurity [4]–[6]. This literature review concentrates on methodologies for recognizing phishing websites, encompassing explainable AI (XAI), machine learning (ML), deep learning (DL) ensemble strategies, and multimodal detection frameworks. The reviewed study incorporates large language models (LLMs) and innovative paradigms, including quantum-enhanced learning, featuring URL-based, visual-based, content-based, and hybrid techniques [12]–[14]. The shift to learning-based methodologies is validated by citing conventional blacklist and rule-based frameworks [1], [7]. The scope is restricted to peer-reviewed academic articles, conference papers, and credible datasets, including PhishTank, Alexa, UNB, UCI, and Kaggle datasets, which are commonly utilized in phishing detection research [4], [8]. The review focusses on studies that look at how well these elements work in real life, how well the model holds up, how fast it runs, and how easy it is to understand [3], [15], [16]. We found the literature we needed for this overview by doing a thorough search of well-known technical databases like IEEE, ACM DL, Springer Link, and Google Scholar [11], [17].

Based on selection criteria, researchers who proposed new and improved detection techniques, offered specific feedback using either standardized results or real-world data sets, addressed important issues like scalability, flexibility, and interpretability, and were clearly relevant to real-time and client-side processing were given high priority. [1], [2], [6], [15], [16], [18]. The type of study that was expressly prohibited relied only on traditional blacklist learning techniques without incorporating [1], [2], [6], [15], [16]. The final set of contributions listed in reflects the transition from inflexible rule-based systems to advanced and adaptive structures that are multimodal and interpretable [11], [13], [14].

A. Rule-Based and Decision Tree Classification Algorithms

Most early phishing detection systems used features that people made by hand and rule-based classification. The research in [1] illustrates this approach by utilizing algorithms like C4.5, RIPPER, PRISM, and CBA to develop a coherent rule-based classifier. The study aims to clarify decision-making by extracting 17 address-bar and domain-specific attributes. The system worked better with only nine key indicators, and it didn't lose much accuracy. Support Vector Machines (SVM), Naïve Bayes, Random Forests, and Decision Trees are some of the most common traditional machine learning classifiers. The study in [7] indicates that Extreme Learning Machines (ELMs) could rapidly categories items utilizing 30 URL-based features. They perform superior to SVM and Naïve Bayes. These methods are easier to use and don't need as much computing power, but they often use static representations of features and have trouble changing to phishing strategies that change quickly. Attackers are less effective when they use real domains that have been hacked or make bad URLs.

B. Ensemble method-based models

Ensemble-based methods are becoming more popular because they work around the problems with single classifiers. The authors in [4] show that XGBoost (boosting) models are better at detecting things after Chi-Square-based feature selection than Random Forest (bagging) models. Ensemble methods make things more adaptable and less variable, but they cost more to compute. Adaptive systems take ensemble learning to the next level. In [19] , Decision Trees, SVM, Random Forest, XGBoost, and MLP are all trained at the same time. After that, the best model is picked out to use. This design makes it easier for people to use a web platform to connect to and analyse data in real time. [2] showed that the CANTINA+ framework has a lot of different parts. An ML engine, a way to find login forms, and filtering based on hashes are all parts of this. There are more than one layer of

protection, which means that there are fewer false positives and more true positives. It is less likely to be hurt by more advanced attacks like XSS and DOM manipulation, though, because it uses static feature sets and hash-based filters.

C. CNN method-based models

Convolutional neural networks (CNNs) are a kind of deep learning method that has been very good at finding phishing attempts. Research including [15], [20], and [21], [16], utilize CNNs to examine URLs, webpage screenshots, and structural arrangements. The multimodal CNN method in [15] achieves 99.67 accuracy through the integration of visual inspection and URL analysis. VisualPhishNet [20] shows how CNN can handle small changes in layout and language by putting phishing pages into three groups based on how similar they look. CNN-based methods are very accurate, but they require a lot of computing power, which makes them less useful in real-time or low-resource settings. Many studies also lack validation in real-time browsing situations and depend on static data [16].

D. Hybrid, Explainable, and Client-Side Approaches

The goal of hybrid models is to find a middle ground between efficiency, performance, and interpretability. The PhishJam system [6] uses ensemble learning to combine manually created URL characteristics with BERT and LSTM embeddings to get very accurate results. Client-side XGBoost-based solutions [3] let you find things in real time on user devices by looking at the properties of URLs and HTML. Explainable AI methods make things clearer. In [14], SHAP is combined with supervised models to identify influential factors that enhance cybersecurity experts' trust and usefulness. But explaining ability frequently results in a more advanced model.

E. Emerging Multi-Modal, LLM-Based, and Quantum Techniques

Recent studies are examining new research. Large language models (LLMs) have shown promise in understanding semantic clues, social engineering patterns, and contextual discrepancies in webpage text [11]. Quantum-enhanced phishing detection using 2-qubit circuits [13] is still mostly experimental and hard to scale, even though it converges faster. The use of multi-modal frameworks that combine textual semantics, visual inspection, and URL analysis is a big step forward. By using the strengths of both CNNs and LLMs, these systems make it harder for zero-day attacks and phishing websites that look real to work [16], [11].

F. LLM and CCN based models

A lot of research relies on static or artificial datasets [20], [8], which makes it hard for them to apply their findings to changing, real-world phishing strategies. This is one of the methodological problems that keep coming up in the field. Another big problem is computational overhead, which is especially bad for CNNs and hybrid deep models. These models need a lot of money and can't be used in real time [15], [20], [21]. There is a trade-off between automation and feature engineering: Deep Learning (DL) models automate feature extraction at the cost of interpretability [14], while traditional machine learning (ML) models require manually created features for interpretability [1]. Moreover, idea drift and adversarial development results in single classifiers and static models experiencing issues with adaptability and drift [7], [4], [2]. Lastly, there is the problem of Incomplete Multi-Modality, which means that many systems don't fully integrate all three types of features; instead, they only focus on one: URL, visual, or content [20], [3], [11]. It's interesting that very few studies actively look into input optimization, which could greatly speed up processing time without lowering accuracy. Finding the shortest URL length or the best resolution for a screenshot are two examples of this.

Although phishing website detection has advanced significantly, there are still a number of important research gaps. Many current methods either utilize lightweight models that are not resistant against complex and constantly changing phishing attempts [7], [21], or they prioritize detection accuracy at the expense of computational performance [15], [20]. There are yet few fully integrated multi-modal detection methods in the literature that simultaneously make use of semantic content understanding, visual webpage characteristics, and URL structure information [3], [11]. Additionally, crucial practical factors are frequently overlooked, including adaptive learning under idea drift, real-time deployment, and systematic input feature optimization [14], [16]. A computationally effective, explainable, and adaptable multi-modal phishing detection approach that combines CNN-based visual feature extraction with LLM-based semantic analysis of URLs and webpage content is desperately needed to close these gaps. In contexts with limited resources, such a framework should be able to allow real-time operation while maintaining high detection accuracy. We recommend further developing the model in subsequent work by creating an automated system to determine the smallest informative webpage screenshot size and the minimum effective URL length necessary for accurate detection. This solution would help implement scalable, real-time phishing detection systems in contemporary online contexts by lowering computing cost while maintaining performance.

III. Methodology

Through our research we developed phishing website detection through a multi-modal learning framework that combines visual and text analysis. The detection task is structured as a binary classification problem, categorizing a website as either legitimate or phishing based on its webpage screenshot and URL. A convolutional neural network (CNN) extracts visual features from optimized webpage screenshots, and a large language model (LLM) extracts semantic and structural patterns from optimized URL representations. Input optimization is done to lower the resolution of screenshots and the length of URLs while keeping detection accuracy and lowering the cost

of computation. The resulting visual and linguistic features are then combined to generate the final classification.

A. Datasets

The Phishing URL Dataset from Hugging Face has about 10,294 URL samples, with 5,074 of them being real URLs and 5,220 of them being phishing URLs. This makes the class distribution almost equal.We splited the URL dataset.80% of them are used for taining, 10% testing and 10% for validation.This gives about 4050 legit URL and 4185 phishing URL for testing, 514 legil URL and 515 phishing URL for validation and at last 510 legit URL and 520 phishing URL for testing. There are 16 attributes in each instance, and the raw URL string is the main feature used to input data into the model. The URLs are marked with a 0 for real and a 1 for phishing, which makes it possible to classify them in two ways. This dataset is great for training and fine-tuning big language models to pick up on the semantic, lexical, and structural features of modern phishing URLs because it is big, varied, and has high-quality annotations.For picture analysis we use a Verified Phishing and Genuine Page Screenshots dataset from Kaggle, it has 2,282 screenshots of web pages that have been rendered. Of these, 1166 are real and 1,116 are phishing pages. The dataset has JPEG and PNG image files of fully rendered web pages from both phishing and legitimate sites. The dataset is splited in 70% for training, 15% validation and 15% for testing. Among the 1166 real website screenshot 806 of them is for training, 180 of them is for validation and the last 180 of them is for testing.For the 1,116 fake website screenshot 763 of them is for training, 169 and 184 of them is for validation and testing. andThe convolutional neural network (CNN) takes in each image as a set of raw pixels. This lets it find visual features on its own. This dataset is great for teaching visual detection models because it lets the CNN learn how to tell the difference between things like page layout, logos, colour schemes, and branding patterns that are often used in phishing attacks that use visual spoofing.

B. Image and URL Preprocessing

For visual data processing, screenshots of web pages with different resolutions are first resized to a standard size. To make sure that all of the images look the same, this is done by down sampling higher-resolution images and up sampling lower-resolution images. Then, all of the screenshots are combined into one image tensor and set to a specific spatial resolution (for example, 224 × 224 pixels with three color channels). This step makes each screenshot the same size so that the convolutional neural network can process them in groups and find features in them. The raw URL string goes through a number of steps to get ready for the language model. The first step is to change all of the characters to lowercase to make sure that all of the URLs are the same. After that, any extra punctuation and special characters are removed. After cleaning, the URLs are split into sequential tokens and either padded or cut down to a set length so that they can be used as input for the LLM's transformer architecture.

C. Visual Branch: CNN-Based Feature Extraction

The visual branch employs a Convolutional Neural Network (CNN1) to extract high-level visual features that represent the webpage's layout, structural patterns, and logo characteristics. This design enables the detection of phishing attacks that rely on visual imitation, where malicious webpages closely resemble legitimate sites. The CNN1 architecture consists of two convolutional layers followed by two max-pooling layers and a final flattening stage.

1) Convolutional Layers: The input webpage screenshot is processed through a sequence of convolutional kernels to generate feature maps. The convolution operation producing the i-th feature map is defined as:

$$Y_i = f(W_i * X) \tag{1}$$

where X denotes the input image, W_i represents the i-th convolution filter, $*$ indicates the convolution operation, and f is the Rectified Linear Unit (ReLU) activation function. Convolutional filters of size 5×5 are employed to to capture localized visual features effectively.

2) Pooling Layers: Max-pooling layers are applied after each convolution to reduce the dimensionality of the feature maps and retain the most significant visual features, enhancing scale invariance.

D. URL Feature Extraction (LLM)

The linguistic branch utilizes a Large Language Model (LLM), such as a fine-tuned BERT or a similar transformer-based architecture, to analyze the URL string. This represents a significant departure from traditional approaches, as the LLM is particularly effective at the following tasks:

- **Lexical Analysis:** Detecting subtle linguistic cues such as character repetition, suspicious keyword placement, and misleading subdomain structures.
- **Contextual Understanding:** Assessing the overall pattern and entropy of the URL string, providing a more sophisticated representation than simple character-level CNNs or rule-based techniques.

The output of the LLM's final hidden layer corresponding to the `[CLS]` token (or an equivalent pooled representation) is extracted and used as a dense URL feature vector, denoted as $\mathbf{V}_{\text{URL}}$.

E. Fusion and Classification

The feature vectors generated by the two processing branches—namely, the visual feature vector $\mathbf{V}_{\text{Visual}}$ obtained from CNN1 and the URL feature vector $\mathbf{V}_{\text{URL}}$ extracted by the LLM—are combined using vector concatenation to form a unified representation:

$$\mathbf{V}_{\text{Combined}} = [\mathbf{V}_{\text{Visual}} \,;\, \mathbf{V}_{\text{URL}}] \tag{2}$$

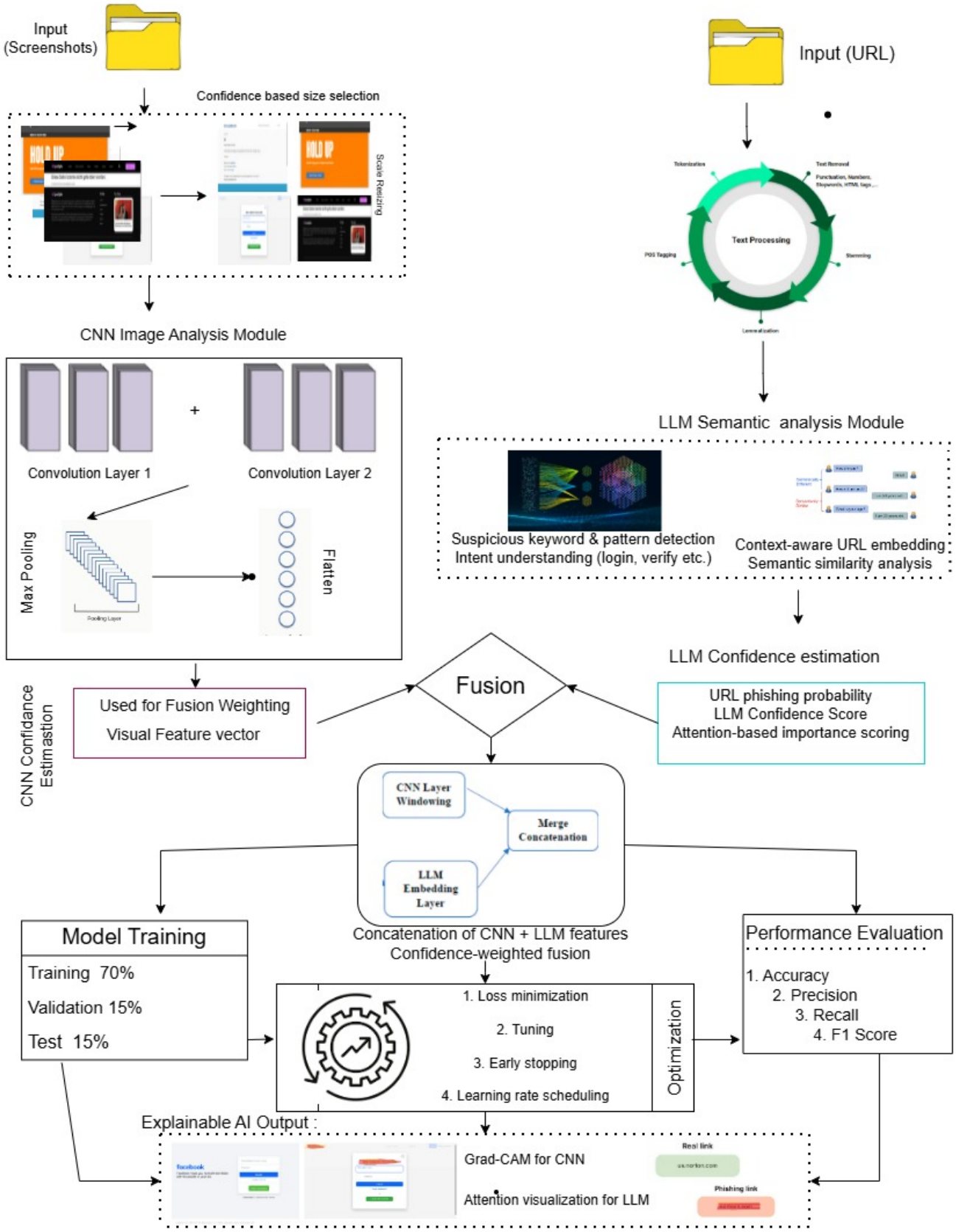


Fig. 1: Multi-Modal Phishing Detection Framework

The combined feature vector is subsequently fed into a fully connected (FC) layer that performs the final classification. A Sigmoid activation function is applied to the FC layer output to compute the predicted probability α that the input webpage is phishing:

$$\alpha \cdot \text{cnn_score} + (1 - \alpha) \cdot \text{llm_score}$$

where α denote the learnable weight matrix and bias term, respectively represents the Sigmoid function. A decision threshold of 0.5 is used to convert the predicted probability into a binary label, such that webpages with $alpha \geq 0.5$ are classified as phishing (1), while those with $alpha < 0.5$ are classified as legitimate (0).

F. Evaluation Metrics

To evaluate the performance of the proposed model, multiple standard classification metrics are employed. Accuracy is used as the primary evaluation metric and is complemented by Precision, Recall, and the F1-score. These metrics are particularly important for phishing detection, which often involves class imbalance and requires careful handling of false positives and false negatives.

1) Accuracy: Accuracy measures the overall correctness of the classification model and is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

2) Precision: Precision quantifies the proportion of predicted phishing webpages that are actually phishing, thereby minimizing false positives and reducing alert fatigue:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4}$$

3) Recall: Recall measures the model's ability to correctly identify phishing webpages, emphasizing the reduction of false negatives and improving threat detection:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5}$$

4) F1-Score: The F1-score represents the harmonic mean of Precision and Recall, providing a balanced measure of classification performance:

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6}$$

Here, TP (True Positive), TN (True Negative), FP (False Positive), and FN (False Negative) are derived from the confusion matrix.

G. Experiment Setup

The proposed phishing detection system was developed in Python utilizing the PyTorch deep learning framework. The Hugging Face Transformers library let us use pre-trained transformer-based language models to look at URLs. We used PyTorch's convolutional neural networks to get visual features. OpenCV and Pillow (PIL) got the pictures ready. Scikit-learn helped us figure out how to test models, and NumPy and Pandas helped us with math and data processing. We used Jupyter Notebook to run and make experiments, and Git to keep track of different versions. This research utilized two publicly accessible datasets. We used the GPT Neo Phishing URL Dataset from Hugging Face for our language analysis.

IV. Result and Analysis

To implement our approach we used Python Programming Language and a dataset containing2282 screenshots and URLs of legitimate and phishing websites. We measured the performance of the proposed method in terms of accuracy, F1 score, precision and recall. The experiment result is shown in Table 3. The results reflect the effectiveness of using CNNs as an automatic way to extract features and carry out a classification task to distinguish between reliable and phishing web pages.

Metric	Value
Accuracy	95.3 %
Precision	98 %
F1 score	95.16 %
Recall	94.76 %

A. Graph

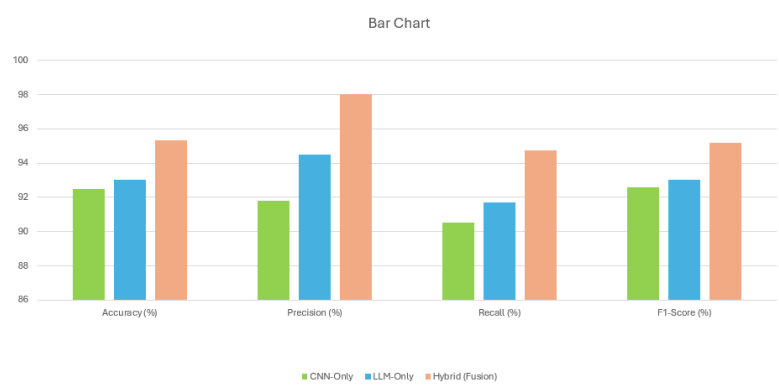
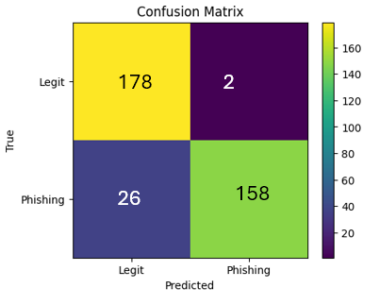


Fig. 2: Performance Analysis Graph

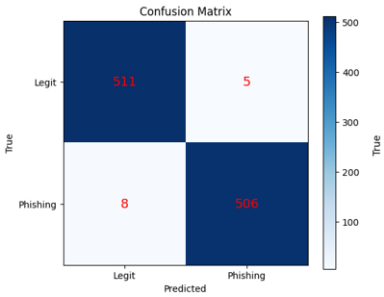
The Hybrid model consistently outperforms both single-method approaches in every evaluated category. The Hybrid model reaches a peak precision of 98.0%, which is much higher than the CNN and LLM models' respective results of 91.8% and 94.5%. The combination of visual and URL detection also significantly improves overall accuracy, which rises to 95.3% for the Hybrid system from 92.5% for CNN-Only and 93% for LLM-Only. Additionally, the Hybrid model maintains high sensitivity with a balanced F1-Score of 95.16% and a recall of 94.76%, demonstrating a strong balance between recall and precision.

B. Confusion Matrix

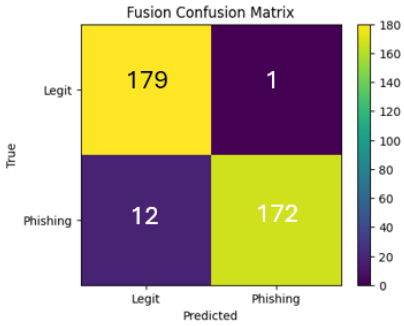
The three confusion matrices show a detailed comparison of the classification performance for the CNN, URL. The CNN confusion matrix shows that out of 180 true legitimate screenshots, it correctly identified 178 legit but misclassified 2 legitimate



(a) CNN Confusion Matrix



(b) URL Confusion Matrix



(c) Fusion Confusion Matrix

Fig. 3: Comparison of Confusion Matrices for CNN, URL, and Fusion Models

cases as phising, resulting in false negative. The URL-based model shows higher processing, correctly identifying 511 legitimate and 506 phishing cases, with only 5 false negative and 8 false positive. At last, the Fusion Confusion Matrix shows the peak performance of the hybrid approach, correctly identifying 179 legitimate sites and 172 phishing sites.

V. Discussion

The key finding from this research is that there is collaboration produced when these visual features are fused through CNN and these lexical features are fused through DistilBERT. The reason is evident from the performance evaluation chart, where the highest accuracy demonstrated by the Hybrid Fusion model is 95.3%.

This is due to the fact that the model possesses the capacity to detect “cross-modal deception”. Alternatively, if the attacker employs a whole new page design in an attempt to deceive visualization detection, DistilBERT recognizes the visual URL deception.Our findings suggest that the model is particularly resilient to clones that are ”pixel-perfect.” We grouped the detections into three levels of difficulty. It proves that a model is not just ”memorizing” colors; rather, it understands the spatial hierarchy of login forms and logo placements. Robustness to Variations: The hybrid system continued to perform well against attackers who introduced small errors, such as changes in the language of the webpage or background banners, which often easily break systems based on template matching.

The URL-only model had a high recall, while the main challenge was represented by the considerable number of false positives due to legitimate subdomains that looked suspicious.In fact, the CNN-only model was highly accurate but with a higher number of False positive, which counts up to 26 cases when it comes to the original layouts. This translates into, the Fusion Model effectively ”cancelling out” these errors to decrease the false positives to a count of 12. However, there were ”hard cases” observed with platforms like Microsoft Office and Dropbox. The errors in these discussions suggest that these platforms have such minimalist designs that the visual ”saliency” is low, making it hard for the CNN to extract unique identifying features with more contextual metadata at hand.

It chose for DistilBERT over the full BERT model and EfficientNet-B0 over heavier architectures such as ResNet-152, among other conscious choices to balance between accuracy and computational efficiency. The results clearly prove that prohibitive hardware is not needed for high-level security. The fast inference times observed indicate that this model is ready to be deployed for real-time browser extensions for proactive defense against ”zero-day” phishing sites that have not yet been added to global blacklists.

VI. Conclusion

The proposed study introduced a new hybrid phishing detector that combined the use of Convolutional Neutral Networks for visual processing and the use of Large Language Models, referred to as DistilBERT, for URL analysis. The combination of the two approaches proved effective in overcoming the constraints of a single model-based security system. The main findings and contributions of this research can be highlighted as follows. Improvement in Accuracy by Fusion both visual and URL feature-detection codes have improved the accuracy of detection by a considerable margin. This is because the proposed system not only relies on image detection but also on URL detection. Experimental results have already revealed

that the proposed system is 95.3% accurate. Resistance against dynamic threats this modern system is highly strong against zero-day attacks as well as advanced versions of phishing attacks that may try to use visual mimicry as well as hidden URLs. Optimal Parameters for Performance it performed exceptionally well in all the parameters of the highest importance. It had a precision of 98%, an F1 score of 95.16%, along with a recall of 94.76%. Moreover, it was observed that while training, for maximizing accuracy, a batch size of 16 was realized as ideal. Real-World Applicability the fact that the system works purely from the URL and the screenshot eliminates any dependencies on the website's language or the content it holds, making it a very good fit for implementing within real-world cybersecurity tools such as browser add-ons or real-time network analyzers. In conclusion, this hybrid solution is a strong, proactive defense tool to protect digital spaces from ever more complex types of fraud.

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